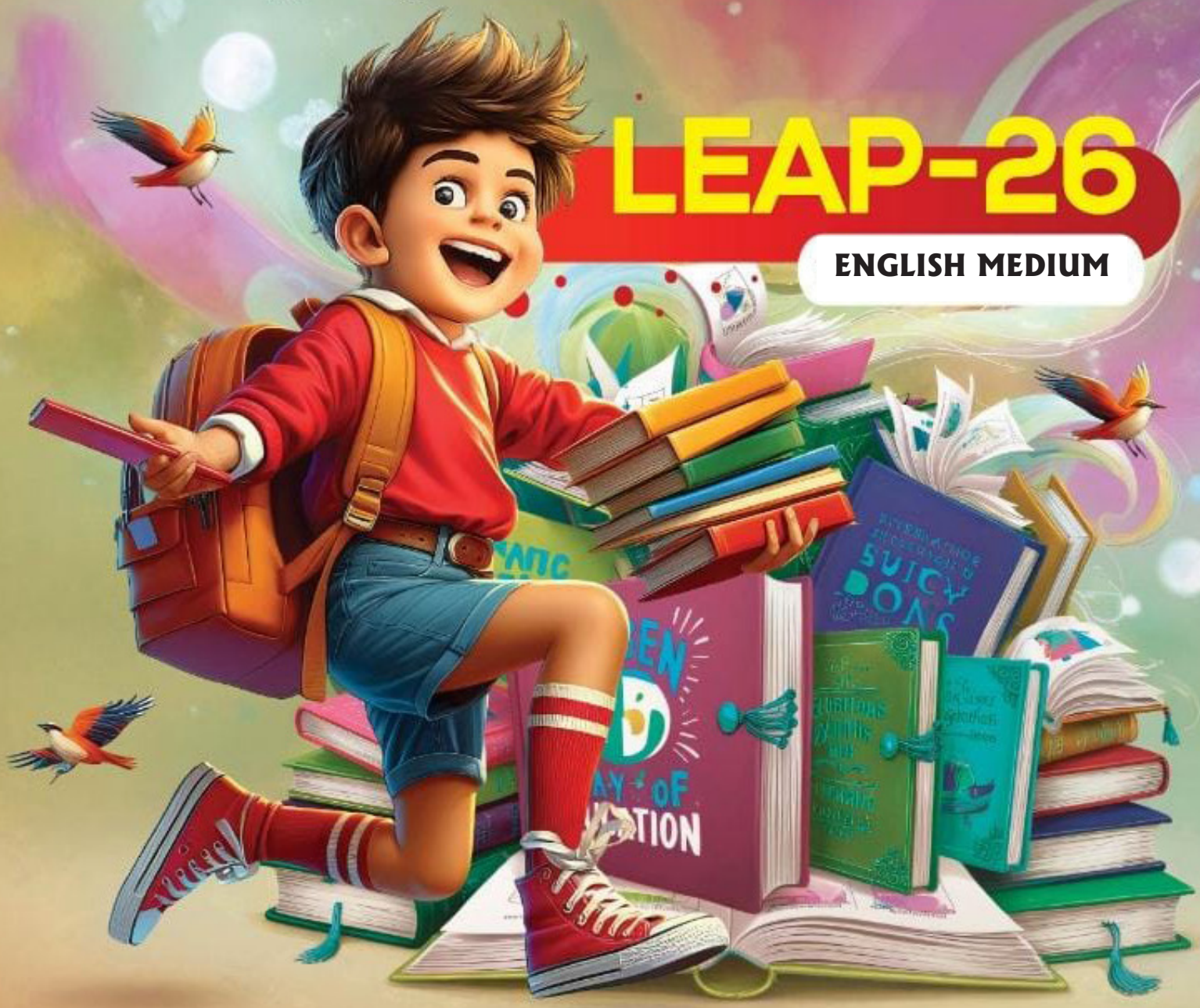
വിജയഭേരി

മലപ്പുറം ജില്ലാ പഞ്ചായത്ത് വിദ്യാഭ്യാസപദ്ധതി

LEAP-26

ENGLISH MEDIUM



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VIJAYABHERI
MALAPPURAM DISTRICT PANCHAYAT
EDUCATIONAL PROJECT

LEAP 26
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അഭിമുഖം

2001–2002 വർഷത്തിൽ മലപ്പുറം ജില്ലയുടെ സമഗ്ര വിദ്യാഭ്യാസ പുരോഗതി ലക്ഷ്യം വെച്ചു മലപ്പുറം ജില്ലാ പഞ്ചായത്ത് ആസൂത്രണം ചെയ്തു നടപ്പിലാക്കിവരുന്ന പദ്ധതിയാണ് വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതി.
കഴിഞ്ഞ 24 വർഷത്തിനുള്ളിൽ ജില്ലയുടെ വിദ്യാഭ്യാസ മുഖച്ഛായ മാറ്റാൻ ഈ പദ്ധതിയിലൂടെ കഴിഞ്ഞിട്ടുണ്ട്. എസ്എസ്എൽസി വിജയശതമാനത്തിലും പ്ലസ് ടു വിജയശതമാനത്തിലും മത്സര പരീക്ഷകളിലും ഏറെ പിന്നിലായിരുന്ന ജില്ല ഇന്ന് സംസ്ഥാനത്തെ മറ്റേത് ജില്ലയോടും കിടപിടിക്കാവുന്ന രീതിയിൽ മാത്രമല്ല, പല മേഖലയിലും അവരെക്കാൾ മുന്നേറാൻ ഇന്ന് ജില്ലയ്ക്ക് കഴിഞ്ഞിട്ടുണ്ട്.
ഏറ്റവും കൂടുതൽ എ പ്ലസ്സുകളുള്ള ജില്ല, ഏറ്റവും കൂടുതൽ കുട്ടികൾ കേന്ദ്ര സർവകലാശാലയിലേക്ക് പ്രവേശനം നേടുന്ന ജില്ല, മെഡിക്കൽ എൻജിനീയറിങ് എൻട്രൻസ് പരീക്ഷകളിൽ മുന്നിൽ നിൽക്കുന്ന ജില്ല ഇന്ന് മലപ്പുറം ജില്ലയാണ്. വിദ്യാഭ്യാസപരമായി പിന്നാക്കം നിൽക്കുന്ന ജില്ല എന്ന അപഖ്യാതി നമ്മളിന് മറികടന്നു.

മറ്റ ജില്ലകളിൽ നിന്ന് ഭിന്നമായി ജില്ലയിലെ എല്ലാ തദ്ദേശ സ്വയംഭരണ സ്ഥാപനങ്ങൾക്കും എംഎൽഎമാർക്കും ഇന്ന് പ്രത്യേക വിദ്യാഭ്യാസ പദ്ധതികളുണ്ട്. ഇത്തരം എല്ലാ പദ്ധതികൾക്കും പ്രചോദനമായത് വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതിയാണ്. സംസ്ഥാനതലത്തിൽ മാത്രമല്ല ദേശീയ തലത്തിൽ പോലും ഈ പദ്ധതി ചർച്ച ചെയ്യപ്പെട്ട വരുന്നു. യൂണിവേഴ്സിറ്റികളിൽ വിജയിഭേരിപദ്ധതി ഗവേഷണ വിദ്യാർത്ഥികൾക്ക് പഠന വിഷയമാണ്.

ഈ പദ്ധതിയുടെ വിജയം മലപ്പുറം ജില്ലാ പഞ്ചായത്തിന്റെ വിജയമല്ല, ഇത് ഏറ്റെടുത്ത അധ്യാപകരുടെയും രക്ഷിതാക്കളുടെയും വിദ്യാർത്ഥികളുടെയും പിന്തുണയാണ് ഈ പദ്ധതിയുടെ വിജയത്തിന്റെ അടിസ്ഥാനം. അതുകൊണ്ടുതന്നെ അവരോടുള്ള കടപ്പാട് ഈ അവസരത്തിൽ പ്രത്യേകം രേഖപ്പെടുത്തുന്നു.

കഴിഞ്ഞ വർഷങ്ങളിലേതുപോലെ ഈ വർഷം നിലവിൽ വന്ന പുതിയ ജില്ലാ പഞ്ചായത്ത് ബോർഡും വിജയഭേരി പദ്ധതി കൂടുതൽ ഊർജ്ജസ്വലമായി മുന്നോട്ടുകൊണ്ടുപോകാൻ തീരുമാനിച്ചിട്ടുണ്ട്. അതിന്റെ ആദ്യപടിയായി ഈ വർഷം എസ്എസ്എൽസി പരീക്ഷ എഴുതുന്ന വിദ്യാർത്ഥികൾക്കും അധ്യാപകർക്കും സഹായകരമായ രീതിയിൽ എല്ലാ വിഷയങ്ങളുടെയും ചോദ്യശേഖരം പുറത്തിറക്കുകയാണ്.

പുതിയ പാഠപുസ്തകങ്ങളും പുതിയ രീതിയിലുള്ള പരീക്ഷ രീതിയുമാണ് ഈ വർഷത്തെ വിദ്യാർഥികൾ അഭിമുഖീകരിക്കേണ്ടത്. അവർക്ക് ഏറെ സഹായകമായ രീതിയിൽ റിവിഷൻ നടത്തുന്നതിനും പുതിയ രീതിയിലുള്ള ചോദ്യങ്ങൾ പരിചയപ്പെടുത്തുന്നതിനുമാണ് ഇത്തരമൊരു ചോദ്യശേഖരം തയ്യാറാക്കിയിരിക്കുന്നത്. എല്ലാ വിഷയങ്ങളുടെയും എല്ലാ പാഠങ്ങളിലുമുള്ള പ്രതീക്ഷിത ചോദ്യങ്ങൾ ഇതിൽ ലഭ്യമാക്കിയിട്ടുണ്ട്. ഈ ചോദ്യ ശേഖരം വിദ്യാർത്ഥികൾക്കും അധ്യാപകർക്കും ഏറെ സഹായകരമായിരിക്കുമെന്ന് പ്രതീക്ഷിക്കുന്നു.

ദൈനംദിന പ്രവർത്തനങ്ങൾക്കിടയിൽ ഇത്തരമൊരു ഭാരിച്ച ജോലി ഏറ്റെടുത്ത് കൃത്യസമയത്ത് പൂർത്തീകരിച്ച് പ്രസിദ്ധീകരിക്കുന്നതിന് രാപ്പകലോടൊപ്പം നിന്ന അധ്യാപകരെ പ്രത്യേകം അഭിനന്ദിക്കുന്നു.

പി എ ജബ്ബാർ ഹാജി
പ്രസിഡന്റ്
മലപ്പുറം ജില്ലാ പഞ്ചായത്ത്

അഡ്വ. എ.പി. സുജിത
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പി കെ അസ്മു
ആരോഗ്യ-വിദ്യാഭ്യാസ സ്റ്റാൻഡിംഗ് കമ്മിറ്റി ചെയർമാൻ

ടി സലിം
കോഡിനേറ്റർ
വിജയഭേരി വിദ്യാഭ്യാസ പദ്ധതി

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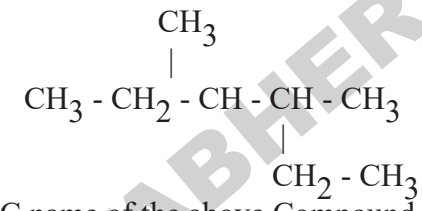
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1 NOMENCLATURE OF ORGANIC COMPOUNDS AND ISOMERISM

1 Marks

1. Two compounds are given.
   (i) $CH_3 - CH_2 - CO - CH_2 - CH_3$
   (ii) $CH_3 - CO - CH_2 - CH_2 - CH_3$
   Statement 1: (i) and (ii) are position isomers
   Statement 2: (i) and (ii)are metamers
   A. Statements 1 and 2 are correct.
   B. Statement 1 is correct, but 2 is incorrect.
   C. Statement 1 incorrect, but 2 is correct.
   D. Statements 1 and 2 are incorrect.
2. Find out the compound which shows both position isomerism & metamerism from those listed below
   (Butane, Butanal, Butanone, Butanoic acid)
3. Statement 1 - The molecular formula of But - 1 - ene and But - 2 - ene are same
   Statement 2 - But -1-ene and But - 2 - ene shows position isomerism.
   a. Statement 1 is correct but 2 is wrong
   b. Statement 1 and 2 are correct
   c. Statement 1 is wrong but 2 is correct
   d. Statement 1 and 2 are wrong
4. 
   IUPAC name of the above Compound is
   a) 3 - methyl-4-ethyl pentane
   b) 4, 3 - Dimethyl hexane
   c) 3, 4 - Dimethyl hexane
   d) 2 - ethyl-3-methyl pentane
5. The word root of the name of an alkane is oct. The total number of hydrogen atom present in one molecule of it is.....

2 Marks

6. Give the structure and IUPAC name of the compound formed by the substitution of one Bromine atom on the 3rd Carbon of Pentane.
7. One molecule of an aromatic hydrocarbon has 6 carbon atoms.
   a) Name the compound
   b) One of the hydrogen atom is replaced by an -OH group, then write the structural formula & IUPAC name of the compound
8. IUPAC name of an alkane is given as 4 - ethyl pentane.
   Draw the structure of the given compound and check wether the given name is correct or not? Justify your answer.

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9. Draw the structure and give the IUPAC names of the compounds from the given hints.
   a. One H atom of C_6H_6 is replaced by -COOH group.
   b. Two H atom of C_2 in butane replaced by two bromine atoms.

3 Marks

10. (i) Write the structural formula of 2,3,3- Trimethylpentane .
    (ii) Write the structural formula of any one of its chain isomer.
    (iii) Write the molecular formula of the compound (i) when the three methyl groups are replaced by three chlorine atoms.
11. An alkane P has 4 carbon atoms in the main chain (No branches)
    Q is a compound which contains 1 carbon & 2 hydrogen atoms more than the compound P
    R is a compound which forms which one hydrogen atom in the 2nd carbon in Q is replaced by a methyl group
    a. Draw the R
    b. Write the IUPAC name for compound Q
    c.Write the molecular formula of compound P
12. A student gives the name 3-ethylpentane to a compound. Draw its structure and verify if the name is correct. If not, write the correct IUPAC name and explain the rule violated.
13. Look at the compounds given in the bracket
    (Methanoic acid, Propane, chloro butane,propyne, Methoxy methane)
    a) Identity the hydrocarbon with triple bond?
    b) which one has position isomer ?
    c) Which one has only one carbon atom ?

4 Marks

14. The molecular formula of an alkane is C_4H_{10}
    (i) Write the structural formula of the compound when a Hydrogen atom in the second carbon of the above compound is replaced by a methyl group.
    (ii) If a double bond is formed between the second (carbon bearing methyl group) and third carbon in the above compound, what will be its molecular formula and structure?
    (iii) Write the IUPAC name of the compound obtained.
15. Write the IUPAC names for the followings
    a. $CH_3 - CH = CH - CH_3$
    b. $CH_3 - CH_2 - CH_2 - OH - CH_3$
    c. $CH_3 - CH_2 - \overset{\overset{O}{\|}}{C} - CH_3$
    d. $CH_3 - CH_2 - \overset{\overset{CH_2-CH_3}{|}}{CH} - CH_3$
16. Write the structutal formula for the following s
    a. Pentan-2-ol
    b. Butan -1-al
    c. Hex-3-ene
    d. 2,3- dimethyl pentane.

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2 CHEMICAL REACTIONS OF ORGANIC COMPOUNDS

1 Marks

1. Statement 1: Aspirin is an Analgesic
   Statement 2: Paracetamol is both an Antiseptic and an Analgesic
   A. Statements 1 and 2 are correct
   B. Statement 1 is correct but 2 is incorrect
   C. Statement 1 is incorrect but 2 is correct
   D. Statements 1 and 2 are I ncorrect
2. Analyse the given statements
   i) Nylon 66 is an additional polymer
   ii) Phenol and formaldehyde are monomers of Nylon 66
   a) statement I is true, statement ii is false
   b) statement i is false, statement ii is true
   c) Both statements are true
   d) Both statements are false
3. The general formula of ester is.....

   O O O O
   ‖ ‖ ‖ ‖
   -C-OH, -C-, R-C-O-, H-C-
4. Analyse the hints and identify the polymer
   Hints: 1. It is a condensation polymer
   2 .It is used in the manufacture of tarpaulin, bottles, fabrics etc.
5. Which of the given compound undergo addition reaction?
   (Hexane, Chloro hexane, Hexyne, Cyclo hexane)

2 Marks

6. $CH_3 - CH_2 - CH_3 \longrightarrow A + CH_2{=}CH_2$
   $A + 2O_2 \longrightarrow B + 2H_2O + Heat$
   Identify A and B
7. A hydrocarbon X contains one carbon and its genera formula is C_nH_{2n+2}.
   When X react with one Chlorine molecule,
   a) Write the products formed here?
   b) What type of chemical reaction is this?
8. Select wrong pairs from the following and correct them
   i) Vinyl chloride : PVC
   ii) Vinyl cyanide: Natural Rubber
   iii) Tetra flouroethene: Orlon
   iv) Ethene : polythene
9. What is meant by:
   (i) Power alcohol
   (ii) Formalin
10. An alkyne X on addition reaction with hydrogen gives an alkene Y which on polymerisation gives polythene.
    (i) Identify X
    (ii) Write the structural formula of the product when Y reacts with HCl

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3 Marks

11. (i) $CH_3 - CH_2 - CH_2 - CH_3 \xrightarrow{heat} CH_4 + A$
    (ii) $A + H_2 \longrightarrow B$
    (a) Identify A and B
    (b) What is the name of the reaction (i)?
    (c) What is the application of such reactions in our day to day life?
12. The Hydrocarbon 'A' with two carbon atoms and triple bond between them undergo addition reaction with HCl to form 'B'. B undergo polymerisation and form C
    (a) Find A, B and C
    (b) Write common name of B
    (c) Write any two uses of C
13 (i) $CH_3\text{-}OH + X \xrightarrow{catalyst} Y$
    (ii) Y is the carboxylic acid used for the preparation of an ester having the jasmine smell.
    (a) Identify X and Y
    (b) Name the ester having jasmine smell
    (c) Which is the alcohol used for the preparation of the above ester?
14. Medicines play a pivotal role in enhancing health and life span
    a. Which category of medicine is used to control microorganisms?
    b. Name a medicine which has anti-blood coagulant property
    c. Write a function of antibiotics in modern method of medical therapy

4 Marks

15. Analyse the given chemical equation $CH_3\text{-}CH_2\text{-}CH_2\text{-}CH_2\text{-}CH_3 \rightarrow CH_2 = CH_2 + ...$
    a) Complete the chemical equation
    b) What type of hydrocarbons are the products of this reaction?
    c) What are the factors affect the products obtained as a result of this reaction?
    d) Write the definition of this reaction
16. Analyse the given chemical equation (A, B and C are not real)
    $2H_2 + CO \rightarrow A$
    $A + CO \rightarrow B$
    $A + B \rightarrow C + H_2O$
    a) Find A, B and C
    b) Write any two uses of A
    c) How can prepare 5-8% B
    d) Name the reaction of the formation of C
17. Ethanol is prepared by the fermentation of molasses. Analyse the given chemical equations
    $C_{12}H_{22}O_{11} + H_2O \xrightarrow{P} C_6H_{12}O_6 + C_6H_{12}O_6$
    $C_6H_{12}O_6 \xrightarrow{Q} R + 2CO_2$
    a) Find P ,Q and R
    b) How can we prepare rectified spirit?
    c) What do you mean by denatured spirit?
    d) What are the components of methylated spirit?
18. (a) How many molecules of chlorine is required to prepare Hexachloroethane from $C_2H_4C_{l2}$
    (b) Write the equations of each step of the above reaction
    (c) Write the IUPAC name of the hydrocarbon formed in each step
    (d) Write the name of such type of reactions

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3 PERIODIC TABLE AND ELECTRON CONFIGURATION

1 Marks

1) Fill in the blanks with correct pair
d block-transition elements
f block-.....

2) Which of the following are incorrect electronic arrangements.

a) $1s^2\ 2s^2\ 2p^6$ b) $1s^2\ 2s^3\ 2p^5$
c) $1s^2\ 2s^2\ 2p^7$ d) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1\ 4s^2$

A- a & b
B- b & c
C- c & d
D- a & d

3) Subshell electronic configuration of X^{2+} ion is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

Statement 1: X is a p block element.
Statement 2: X is present in 4th period of the periodic table.
Statement 3: Oxide of X is represented as X_2O

Which of the following is correct regarding the given statements?

A Statement 1,2,3 are not correct.
B Statement 1 is correct,but statements 2 and 3 are not correct.
C. Statement 2 is correct,but statements 1 and 3 are not correct.
D. Statement 1and 3 are correct ,but statement 2 is not correct.

4) How many values will be there for m when l=3 ?

2 Marks

5) a) Find out the oxidation state of Mn in MnO_2? (oxidation state of oxygen is -2)
b) Write the subshell electronic configuration of that Mn ion?

6) The electronic configuration of an element is given as $[Ar]3d^{10}4s^2$.
Write down the electronic configuration of the element which is present in the same period, left to this element in the periodic table?

7) When $K_2Cr_2O_7$ is converted into K_2CrO_4, the change in oxidation number of chromium is

a) 6 b) 0 c) 4 d) 3

b) Transition elements show variable oxidation states. Give reason?

8) Match the following

A	B
a) $1s^2\ 2s^2\ 2p^5$	i) Group 13 element
b) $1s^2\ 2s^2\ 2p^6\ 3s^2$	ii) Transition element
c) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$	iii) Alkaline earth metal
d) $1s^2\ 2s^2\ 2p^1$	iv) Most reactive non metal

Choose the correct one from the options given below

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	a)	b)	c)	d)
A	ii	iii	iv	i
B	iv	iii	ii	i
C	iii	iv	i	ii
D	ii	iv	i	iii

3 Marks

9) Subshell electronic configuration of two elements are given below.

A=[Ar]$3d^1 4s^2$ B=[Ar]$3d^{10} 4s^2$

a) To which block these elements belongs to?

b) Write down any two characteristics of this block elements?

10) An atom has 3 shells and 6 electrons in its outermost shell. Find the following.

a) Subshell electronic configuration

b) Number of s electrons?

c) Period and group

11) The atomic numbers of elements A, B, C are Z-1, Z and Z+1 respectively.
B is an inert gas with atomic number 10.Answer the following

a) Predict the group of A and B?

b) Which element has highest ionisation enthalpy?

c) Which element has lowest ionisation enthalpy?

12) The last 7 electrons of an element X enters into 3d subshell.
Find out the following in relation to X?

a) Complete subshell electronic configururation?

b) Group and period

c) Block

4 Marks

13) The position of two elements A and B in the periodic table is given below

A-Period-2, Group-16

B-Period-3, Group-1

a) Write down the subshell electronic configuration of A and B?

b)Write the chemical formula of the compound obtained by combining A and B?

c)Write the values of n and l of electrons in the outermost subshell of A?

14) a)Which of the following are main block elements
(s and p block,d and f block,s and f block, p and d block)
b)Is ionisation enthalpy lower for s block elements or p block elements?
c) Justify your answer?

15) Some statements about f block elements are given below.

1) Rare earth elements consist of 18 elements including 15 elements of lanthanoids along with scandium and yttrium.
2) Actinoids are radio active elements
3) Elements in 6th and 7th period are actinoids and lathanoids respectively.
4) Neodymium is used for making strong magnets.

a) Identify the wrong statements.

b) Rewrite the wrong statements by correcting it?

16) A few subshells are given below

i) 3d ii) 2p iii) 1s iv) 4f

a) Find the l values of each subshell?

b) Find n+l values and arrange the given subshells in the increasing order of their energies?

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4 GAS LAWS AND MOLE CONCEPT

1 Marks

1) Read the statement and find which one is true

A- The lowest temperature of a gas can reach at 273.15^{o}C

B- Based on this ,The kelvin scale is used for measuring tempersture is developed

a) A and B correct
b) A and B wrong
c) A is correct B is wrong
d) A is wrong B is correct

2) (A) statement: The kinetic energy of the molecules before and after the collision will be same
(R) Reason:The average kinetic energy of the gas molecules is directly propotional to it's temperature

a) A and R correct
b) A Wrong R Correct
c) R is the correct explanation of A
d) R is not the correct explanation of A

3) if the number of atoms in 2 moles of hydrogen is equal to that in 1 mole of oxygen,then their volume at STP will be .....?

2 Marks

4) Identify the main postulates of kinetic molecular theory

a) The volume of gas molecules is negligible in comparison with the total volume of the gas
b) The volume can be reduced by reducing the distance between the gaseous molecules
c) The average kinetic energy of the gas molecules is inversly propotional to it's temperature
d) The attractive force of gas molecule is very high.

5) A Gas kept in a cylinder having 884 L at 27°C with 1 atm pressure

a)Calculate the number of moles present?
b)Calculate the mass of CO_2 which is equal to the no of Molecules present in sample

6) Observe the graph of a Gas Molecule which contain P1,P2 pressure,V Volume, Temperature at 0°C

a) find the value of X
b) write the name of X

3 Marks

7)Observe the pressure-volume graph

a) If the pressure is 4 atm, then find it's volume?
b) which gas law is Applied here?

8) Examine the sample given

a) 128 g O_2
b) 88 g CO_2
c) 4 X 6.022 X 10^{23}

1) same sample contain same volume?
2) How many more grams of the sample should be taken to make the unequal gas sampling in equal?
3) Find the number of atoms in 88g CO_2?

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9) Analyse the situation and answer the following question

1) Liquid ammonia is kept in water for a short time before opening

2) At the same temperature and pressure the number of molecules in two cylinders filled with O_2 and N_2 Gases is the same

a) which gas laws are applied in both situations?

b) which one is match for avogadro's law

4 Marks

10) complete the table

Substance	Molar mass	Given mass	Number of moles	Number of atoms
H_2O	18 g	.....a.....	10	.....b.....
NaCl	58.5 g	.....c....	.....d.....	$20X6.022X10^{23}$
NO_2	.....e....	92	2	.....f.....

11) Observe the chemical equation

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

A) What is the oxygen required to react with 16g methane

B) How many moles of CH_4 Required to produce 264g CO_2

C) Find the volume of CO_2 produced by the complete combustion of 16g of CH_4

D) How Many water molecules required to produce when 320 O_2 molecules react with CH_4

12) The process of neutralising the acidic waste discharged from industrial factories is given

$$H_2SO_4 + Ca(OH)_2 \longrightarrow CaSO_4 + 2H_2O$$

a) Calculate the amount of lime in kg required to neutralise 1960g H_2SO_4

b) Calculate the amount of H_2SO_4 required to neutralise 720kg of lime

c) calculate the number of atoms present in 980g of H_2SO_4

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5 ELECTROCHEMISTRY

1 Marks

1. During the electrolysis of NaCl solution, the product obtained at anode is
   (H_2gas, Cl_2gas, O_2 gas, Sodium metal)
2. Statement 1: Galvanic cell is a device used to produce electical energy from chemical reactions.
   Statement 2: In Galvanic cells, oxidation is take place at cathode and reduction is take place at anode
   a. Statement 1 & 2 are correct
   b. Statement 1 & 2 are not correct
   c. Statement 1 correct and 2 not correct.
   d. Statement 1 not correct and statement 2 correct
3. Which of the following metal can displace Zinc metal from its solutions.
   (Lead, Copper, Silver, Magnesium)
4. Pick out the correct statement from the followings
   1. Electrolysis is the procesd of convering electrical energy in to chemical energy.
   2. During electrolysis, cations moves in to the anode
   3. During electrolysis of Water, Hydrogen gas produced at cathode
   4. During eectrolysis of molten NaCl, Hydrogen gas is produced at cathode.
5. Statement 1: Fuel cells are used in appolo space programmes.
   Statement 2: The byproduct of Fuel cell is water, which is used for drinking purpose of astronauts
   a. Statement 1 & 2 are correct
   b. Statement 1& 2 are not correct
   c.Statement 1 correct and 2 not correct.
   d. Statement 1 not correct and statement 2 correct

2 Marks

1. Consider the displacement reaction given below.
   $2Mg + CuSO_4$ ---------> $MgSO_4 + Cu$.
   a. Which metal is oxidised here?
   b. Write the equation for oxidation.
2. The electric current is produced as a result of redox reaction tsking place in Galvanic cell
   a. Draw the picture of Daniel cell
   b. Mark the flow of current in Daniel cell.
3. Arrange the following electrode pairs in the order of cell voltage.
   Al- Cu, Cu - Ag, Ag- Mg, Zn-Cu.
   b. Write the direction of flow of electrons in each cell
4. Match the following

A	B
a. Mercury cell	1. Fuelcell
b. Lithium ion	2 Salt bridge
c. Methane	3. Primary cell
d. Galvanic cell	4. Secondary cell

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5. Identify the oxidant and reductant in the given reaction

$2Ag(NO_3) + Pb \longrightarrow Pb(NO_3)_2 + 2Ag$.

3 Marks

1. Electrolysis is a process by which an electrolyte undergo chemical changes by the passage of electricity
   a. Write the product obtained at anode and cathode during electrolysis of NaCl solution
   b. The solution around cathode turns pink colour when drops of phenolphthalein is added in to it. why?
2. Some metals are given
   (Pb, Cu, Na, Zn, Mg, Ag)
   a. Which of these metals can displace Hydrogen from HCl?
   b. Justify your answer.
3. a. Define electroplating
   b. An iron ring is electroplated with copper metal.
      1. Which metal is taken as anode?
      2. Which metal is taken as cathode?
      3. Name the electrolyte.
4. Write any three uses of electroplating
5. The batteries used in clocks are dead after their usage, while the batteries used in Vehicles are not dead after their usage.
   Are you agree with these statement?
   If so, justify your answer.
   If not, justify your answer

4 Marks

6. A Copper ring is electroplated with gold metal.
   a) Which metals are taken as anode and cathode.
   b) Name the electrolyte .
   c) Write the chemical equations of anode and cathode.

7. In a galvanic cell electrode A is immersed in $ZnSO_4$ solution. An electrode Fe is immersed in solution B. A voltmeter is connected with the electrodes by using a copper wire.The solutions are linked by a salt bridge.
   a) Find A and B
   b) What will be the direction of electron flow.
   c) Formulate the equation for the chemical reactions at anode and cathode.

8. An iron rod is immersed into the copper sulphate solution in a beaker.
   a) What are the differences between the color of iron rod and $CuSO_4$ solution?
   b) Write the equations of oxidation and reduction reactions.
   c) What will be the change if iron rod is replaced by silver rod? What is the reason?

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6 METALS

Write the answers for questions 1 and 2 each carries 1 mark

1. Match the following.

Extraction method of Ores Metals

a. Electrolysis	i. Mercury
b. Reduction by Carbon	ii. Sodium
c. By controlled heating	iii. Silver
d. Occur more or less in the free state	iv. Zinc

[a. a-ii, b-iii, c-i, d-iv b. a-ii, b-i, c-iv, d-iii c. a-ii, b-iv, c-iii, d-i d. a-ii, b-iv, c-i, d-iii]

2. Match the following

Minerals / Ore Chemical formula

a. Haematite	i. SnO_2
b. Calamine	ii. $ZnCO_3$
c. Cassiterite	iii. Na_3AlF_6
d. Cryolite	iv. Fe_2O_3

[a. a-iv, b-iii, c-i, d-ii b. a-iv, b-ii, c-i, d-iii c. a-iv, b-ii, c-iii, d-i d. a-iii, b-i, c-iv, d-ii]

3. Match the following:

Minerals / Ore Chemical formula

a. Tinstone	i. Al_2O_3
b. Magnetite	ii. SnO_2
c. Bauxite	iii. Cu_2O
d. Cuprite	iv. Fe_3O_4

[a. a-iii, b-iv, c-i, d-ii b. a-ii, b-iv, c-i, d-iii c. a-ii, b-iv, c-iii, d-i d. a-iii, b-i, c-iv, d-ii]

4. Some statements related to concentration of ores are given.

Statement 1: Magnetic separation is a method of concentration of magnetic ore or gangue.

Statement 2: Tin stone is concentrated from Iron tungstate by magnetic separation.

Statement 3: Tin stone, an ore of tin, has no magnetic properties.

a. Statements 1 and 2 are correct. Statement 3 is incorrect.
b. Statements 1, 2 and 3 are correct.
c. Statements 2 and 3 are correct. Statement 1 is incorrect.
d. Statements 1 and 3 are correct. Statement 2 is incorrect.

5. Statement 1: Compounds of highly reactive metals are more stable.
Statement 2: Metals with relatively moderate reactivity also form stable compounds.
Statement 3: Compounds of less reactive metals are less stable.

a. Statements 1 and 2 are correct. Statement 3 is incorrect
b. Statements 2 and 3 are correct. Statement 1 is incorrect.
c. Statements 1, 2 and 3 are correct.
d. Statements 1 and 3 are correct. Statement 2 is incorrect.

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6. Some statements related to minerals are given.
   Statement 1: There may be different minerals of a metal.
   Statement 2: All minerals of a metal are ores of that metal.
   Statement 3: All ores of a metal are minerals of that metal.
   a. Statements 1 and 2 are correct. Statement 3 is incorrect.
   b. Statements 1, 2 and 3 are correct.
   c. Statements 2 and 3 are correct. Statement 1 is incorrect.
   d. Statements 1 and 3 are correct. Statement 2 is incorrect.
7. Different methods are used for the concentration of ores.
   a. Which method is used for the concentration of sulphide ores?
   b. What is the reason for choosing this method?
8. Charge is a mixture added to the blast furnace in the production of iron.
   a. What are the substances contained in the mixture charge?
   b. The name given to the part where the charge is added into the blast furnace?
9. In some cases, alloys are used more than metals.
   a. Write the name of the important alloy of zinc metal?
   b. What are the processes used in the conversion of concentrated zinc ore into zinc oxide?
10. Steel is an important alloy of iron.
   a. Which alloy steel is known as Electrical steel.
   b. Name the alloy steel which is used for making railway tracks.
11. The equation related to the preparation of Zinc metal is given.
   $ZnO + C + heat \longrightarrow Zn + CO$
   a. Why is coke / Carbon used as a reducing agent, instead of electricity in the extraction of zinc?
   b. Name the purification process of zinc. Give the reason for using this method.
12. Different methods are used for the concentration of ores.
   a. Which method is used for the concentration of oxide ores?
   b. What is the reason for choosing this method?
13. Some details related to the industrial preparation of a metal is given.
   i. The ore is treated with hot NaOH solution.
   ii. Electricity is used as reducing agent for the electrolysis of this metal.
   a. What metal is the information given about?
   b. Which mixture is used as an electrolyte when the electrolysis of this metal?
   c. Write the equation for the reaction taking place at cathode.
14. The equations for some of the reactions taking place in a blast furnace are given below.
   $C + O_2 \longrightarrow CO_2$, $CO_2 + C + heat \longrightarrow 2CO$, $CaCO_3 \longrightarrow CaO + CO_2$
   $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$ $\longrightarrow CaO + SiO_2 \longrightarrow CaSiO_3$

   a. What is the function of lime stone in blast furnace.
   b. Which of these is the equation related to the reduction of haematite?
   c. Which substance is act as a reducing agent of haematite? How is it formed?
15. The equation related to the production of copper metal is given.
   $2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$
   a. What is the reason why reducing agent is not used in this activity?
   b. What is Blister Copper? Why it is called so?
   c. Although copper is a metal with low reactivity, what is the reason for using electrolysis for refining copper?

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16. Two situations are given:

Situation 1: Iron nail is wrapped in copper wire.

Situation 2: Iron nail is wrapped in magnesium wire.

a. In which situation does iron rust most quickly?

b. Iron is preserved in some situations. What is the name given to this method of preserving metals?

c. What is the reason for saying that it is not suitable to make devices by combining aluminium wires and copper wires?

17. Pig iron, cast iron, and wrought iron are different types of iron.

a. What is the main difference between these?

b. Which of the iron can be be cast as desired ?

c. How do you make steal from iron?

18. Metal A can displace metal B from its solution. Metal C can displace metal A from its solution. (Symbols are not real)

a. Arrange these metals in decreasing order of reactivity.

b. If a galvanic cell is made using metals B and C, which metal will be its anode?

c. If the valency of A is +2 and the valency of B is +3, write the redox reaction in the galvanic cell A - B.

d. Which two metals among these will give the maximum cell emf?

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7 SOME COMPOUNDS OF INDUSTRIAL IMPORTANCE

1 Marks

1. Match the following.

A	B
a. HCl burner	i. Vanadium pentoxide
b. Sodium hydroxide	ii. Ostwald process
c. Nitric acid	iii. Deionised water
d. Sulphuric acid	iv. membrane cell

[a. a-iii, b-ii, c-i, d-iv b. a-iii, b-i, c-iv, d-ii c. a-iii, b-iv, c-ii, d-i d. a-iii, b-iv, c-i, d-ii]

2. The equations of a reversible reaction at equilibrium is given.

$2SO_2 + O_2 \leftrightarrows 2SO_3$ + heat

Statement 1: In this chemical reaction, the forward reaction increases as the pressure increases.

Statement 2: The rate of forward reaction increases as temperature increases.

Statement 3: Forward reaction increases when SO_3 is removed.

Statement 4: Forward reaction increases when a suitable Catalyst is added.

a. Statements one and two are correct. Statements three and four are incorrect.

b. Statements one and three are correct. Statements two and four are incorrect.

c. Statements one, two, three and four are incorrect.

d. Statement one is correct, Statements two, three and four are incorrect.

2 Marks

3. Basicity is the number of hydrogen ions that an acid molecule can donate.

a. How many H^+ ions are liberated when one molecule of sulphuric acid ionizes?

b. Complete the given ionization equation.

$H_2C_2O_4$ ——> ..... + ...

4. Basicity is the number of hydrogen ions that an acid molecule can donate.

a. What is the basicity of phosphoric acid?

b. Complete the ionization equation and find the compounds x and y.

i. H_3PO_4 ——> .....x..... + H_2PO_4

ii. H_2PO_4 ——> H^+ + ...y.....

5. The equation for the reaction that takes place when sulphuric acid is added to sugar is given below.

$C_{12}H_{22}O_{11}$ + Con. H_2SO_4 ——> 12C + 11 H_2O

a. Which chemical property of sulphuric acid is used here?

b. Write a situation where sulphuric acid is used as a drying agent.

6. Take some ammonium chloride in a test tube and add sodium hydroxide solution to it. Heat the test tube gently.

a. What will be the colour of the solution, when the gas produced in this reaction is passed through water with phenolphthalein added?

b. What can be observed when a glass rod dipped in HCl is placed above the test tube?

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3 Marks

7. The equations of some reversible reactions at equilibrium are given.

i. $SO_3(g) \leftrightarrows SO_2(g) + O_2(g)$

ii. $2NO(g) + O_2(g) \leftrightarrows 2NO_2(g) + heat$

iii. $H_2(g) + I_2(g) \leftrightarrows 2HI(g)$

a. Which of the following reaction is not influenced by pressure?

b. What change in temperature should be made to increase the amount of product in the second reaction?

c. What happens when a catalyst is added to a reversible reaction?

4 Marks

8. The reactor used for the preparation of HCl is known as HCl burner / HCl oven.

a. What is the reason for this name?

b. What are the reactants required for the production of HCl?

c. What is the use of aquaregia?

d. How to convert HCl obtained from an HCl oven into concentrated HCl?

9. Some information related to the industrial preparation of sodium hydroxide is given.

* Brine is the solution used to make sodium hydroxide.

* In modern industries, membrane cell is used for the electrolysis of brine.

a. What is the name of the industrial preparation of sodium hydroxide?

b. How to prepare brine solution?

c. What gases are released during electrolysis?

d. Write the equation for the reduction reaction that takes place in the right chamber of the membrane cell.

10. Some of the elements/nutrients required by plants are given below.

Ca N P K Mg S Fe Mn C H O

a. Which of these are secondary nutrients?

b. What is the difference between primary nutrients and secondary nutrients?

c. How plants get natural nutrients?

d. Is it necessary for watering after fertilization? Why?

11. Salts exhibit acidic, basic, or neutral properties when dissolved in water.

a. Find the basic salts from the given compounds.

NH_4Cl, Na_2CO_3, NaCl, $(NH_4)_2SO_4$, CH_3COONa

b. How to prepare neutral salt?

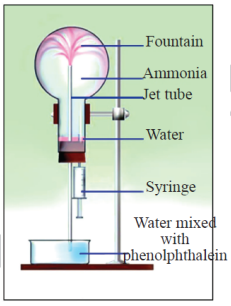
c. What is mean by salt hydrolysis?

d. What acid and base should be used to make sodium carbonate [Na_2CO_3] salt?

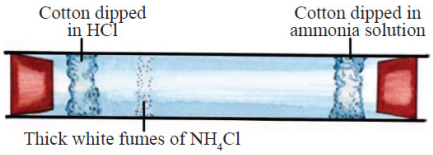
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12. related to the laboratory preparation of Ammonia.
    (a) Which are the reactants used for the preparation of ammonia?
    (b) The ammonia gas produced is collected by inverting the gas jar. What is the reason for this?
    (c) Which substance is used (in the drying tower) to remove the moisture content from ammonia?
    (d) Concentrated Sulphuric Acid (H_2SO_4) is not used to dry ammonia. Why?
13. The experiment to prove the solubility of ammonia is given below. Observe Figure and answer the questions.



    (a) Which observation in the experiment proves that ammonia is a highly soluble gas in water?
    (b) If phenolphthalein is added to the water entering the flask, what color change will occur in the water?
    (c) What property of ammonia is indicated by this color change?
    (d) What is the product formed when ammonia dissolves in water? Write its chemical equation.
14. A cotton plug dipped in ammonia solution is placed at one end of a glass tube, and a cotton plug dipped in Hydrochloric Acid (HCl) is placed at the other end.



    (a) Which chemical is responsible for the formation of white, dense smoke inside the tube?
    (b) Write the equation for this chemical reaction.
    (c) What happens if the product formed here is heated?
    (d) To which category does this product belong? (Acid / Base / Salt)
15. Haber process is the industrial method for the preparation of ammonia.
    (a) Which catalyst is used in this process?
    (b) What is the optimum temperature used in this process to obtain more ammonia?
    (c) What is the difference between 'Liquor ammonia' and 'Liquid ammonia'?
    (d) Write the chemical equation for the Haber process.
16. Consider the reaction

$$N_2(g) + 3H_3(g) \longrightarrow 2NH_3(g) + Heat$$

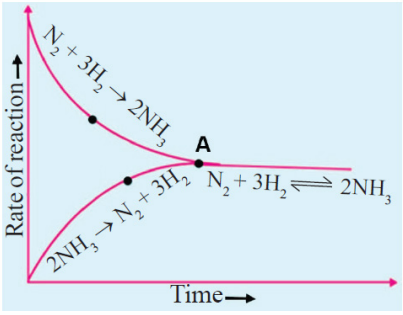
    (a) Is this an Exothermic or an Endothermic reaction?
    (b) What happens if the concentration of nitrogen is increased in this reaction?
    (c) What change will occur in the rate of the forward reaction if ammonia (product) is removed?
    (d) What change will occur in the amount of the product if the pressure is increased in this reaction?
17. Answer the questions related to Le Chatelier's Principle.
    (a) State Le Chatelier's Principle.
    (b) In the reaction

$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$, will the amount of SO_3 increase or decrease if the pressure is increased?

    (c) Does pressure have an influence on the reaction $H_2(g) + I_2(g) \longrightarrow 2HI(g)$?
    (d) If the answer to (c) is "No," what is the reason?

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18. Analyze the graph of a reversible reaction reaching chemical equilibrium.



(a) At the beginning of the chemical reaction (time = 0), which reaction has a higher rate? (Forward reaction / Backward reaction)
(b) What change occurs in the concentration of the reactants over time?
(c) What is Chemical Equilibrium?
(d) Why is chemical equilibrium referred to as a 'Dynamic equilibrium'?

19. Explain the influence of Catalysts on chemical reactions.
(a) How does adding a catalyst affect the rates of the forward and backward reactions in a reversible reaction?
(b) What is the main benefit of using a catalyst?
(c) Which catalyst is used in the Contact Process for the manufacture of Sulphuric Acid?
(d) Iron is used as a catalyst in the preparation of ammonia (Haber process). Then which catalyst is used in the manufacture of Sulphuric Acid?

20. Manufacture of Sulphuric Acid, known as the 'King of Chemicals.'
(a) What is the name of the industrial process used to manufacture Sulphuric Acid?
(b) What is the substance formed by dissolving Sulphur Trioxide (SO_3) in concentrated Sulphuric Acid in this process?
(c) Write the chemical formula for Oleum.
(d) Why is Sulphuric Acid not prepared by directly dissolving Sulphur Trioxide in water?

21. Physical properties of Sulphuric Acid.
(a) Is the density of Sulphuric Acid less than or greater than that of water?
(b) Why is it not advisable to add water to concentrated Sulphuric Acid to dilute it?
(c) What is the correct method for diluting the acid?
(d) Is the dissolution of Sulphuric Acid in water an Exothermic reaction?

22. Dehydrating property of Sulphuric Acid.
(a) What happens when concentrated Sulphuric Acid is added to sugar?
(b) What is the black substance that remains after this reaction?
(c) What color change occurs when concentrated Sulphuric Acid is added to blue Copper Sulphate crystals?
(d) What is Dehydration?

23. Other chemical properties of Sulphuric Acid.
(a) In the reaction $C + 2H_2SO_4 \longrightarrow CO_2 + 2SO_2 + 2H_2O$, which property (Oxidizing/ Reducing) is exhibited by Sulphuric Acid?
(b) The ability to produce Hydrochloric Acid (HCl) by reacting with chloride salts indicates which special property of Sulphuric Acid?
(c) Sulphuric Acid is also a Drying agent. Give an example for this.
(d) In the reaction $Cu + 2H_2SO_4 \longrightarrow CuSO_4 + SO_2 + 2H_2O$, which substance is reduced?

24. Topic: Acid, Base, SaltBasicity of Acids.
(a) What is the 'Basicity' of an acid?
(b) What is the basicity of Hydrochloric Acid (HCl)?
(c) What type of acid is Sulphuric Acid (H_2SO_4)? (Monobasic / Dibasic)
(d) Write the ionization equation for Phosphoric Acid (H_3PO_4).

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26. Arrhenius Theory.
    (a) According to Arrhenius Theory, which ion is released when acids dissolve in water?
    (b) According to Arrhenius Theory, which ion is released when bases (alkalis) dissolve in water?
    (c) Why is Acetic Acid (CH_3COOH) called a 'Weak acid'?
    (d) Give an example of a Strong base.
27. Preparation of Sodium Hydroxide (NaOH).
    (a) What is the name of the industrial process used to manufacture Sodium Hydroxide?
    (b) What is used as the electrolyte in this process?
    (c) Which gas is obtained at the Anode?
    (d) Which gas is obtained at the Cathode?
28. Salts and Hydrolysis.
    (a) Is Sodium Chloride (NaCl) acidic, basic, or neutral?
    (b) What color change occurs if blue litmus paper is dipped in an Ammonium Chloride (NH_4Cl) solution?
    (c) What will be the nature of the salt formed by the reaction between a weak acid and a strong base? (eg. Na_2CO_3)
    (d) What is Salt hydrolysis?
29. Uses of products from the Chlor-Alkali Process.
    (a) Which alkali is obtained through the Chlor-Alkali process?
    (b) Write one use of Sodium Hydroxide.
    (c) Write one use of Chlorine gas obtained in this process.
    (d) Which products from this process are used to manufacture Hydrochloric Acid (HCl)?
30. Topic: Fertilizers - Nutrients required for plant growth.
    (a) Which are the 'Primary nutrients' (NPK) required by plants in large quantities?
    (b) Give two examples of Secondary nutrients.
    (c) What are the nutrients required by plants only in very small quantities called?
    (d) Why are Carbon, Hydrogen, and Oxygen called 'Natural nutrients'?
31. Classification of Fertilizers.
    (a) Write one example of a Nitrogen fertilizer.
    (b) What is the function of nitrogen fertilizers in plants?
    (c) Write one example of a Potash fertilizer.
    (d) How do Phosphate fertilizers help plants?
32. Quality of Fertilizers.
    (a) It is mandatory for fertilizers to be soluble in water. Why?
    (b) Should the use of fertilizers cause a change in the pH value of the soil?
    (c) What are NPK fertilizers or Mixed fertilizers?
    (d) Is the fertilizer Ammonium Sulphate ($(NH_4)_2SO_4$) an acidic salt or a basic salt?
    Would you like me to answer any of these chemistry questions now?

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